BEING FINITE LOCALLY FREE, WITH OR WITHOUT PRESCRIBED DEGREE, IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Finite locally free equals finite, flat and locally of finite presentation. Morphisms, Lemma morphisms-lemma-finite-flat, invoked at proof lines 6451-6453. Splits the first property into three properties whose fpqc locality is already known.
- Finiteness, flatness and local finite presentation are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-finite, lemma-descending-property-flat and lemma-descending-property-locally-finite-presentation, invoked at proof lines 6454-6457. Gives the first assertion.

Source reference. Original lemma 02LA: sources/stacks-descent/pinned-original-source.tex, source lines 6319–6319. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02L2: sources/stacks-descent/pinned-original-source.tex, source lines 6050–6050. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KY: sources/stacks-descent/pinned-original-source.tex, source lines 5984–5984. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KB: sources/stacks-chow/tags.txt, source lines 3366–3366. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is finite locally free" is fpqc local on the base. Let $d \geq 0$. The property $\mathcal{P}(f) =$ "f is finite locally free of degree d" is fpqc local on the base.*

Proof. Being finite locally free is equivalent to being finite, flat and locally of finite presentation (Morphisms, Lemma 02KB). Hence this follows from Lemmas 02LA, 02L2, and 02KY. If $f : Z \to U$ is finite locally free, and $\{U_i \to U\}$ is a surjective family of morphisms such that each pullback $Z \times_U U_i \to U_i$ has degree d, then $Z \to U$ has degree d, for example because we can read off the degree in a point $u \in U$ from the fibre $(f_*\mathcal{O}_Z)_u \otimes_{\mathcal{O}_{U,u}} \kappa(u)$. □